Kernix 0.12.0 / Performance brief

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

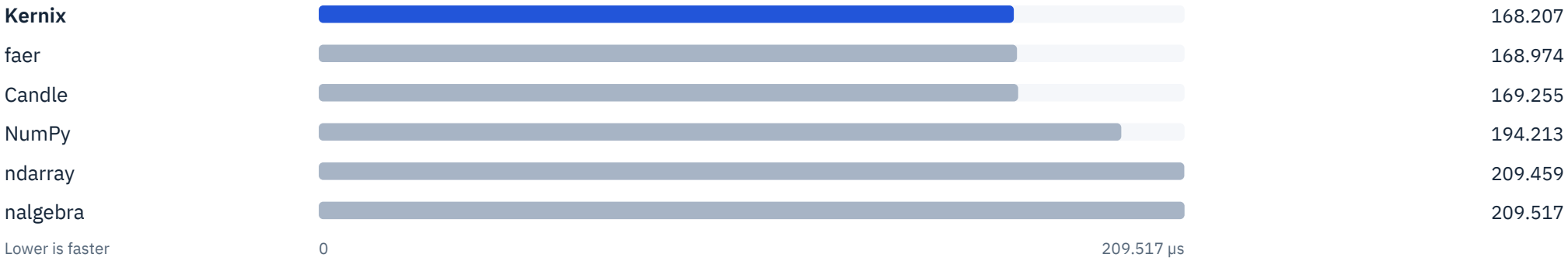
51 workloads

170 observations

Source precision retained

Matrix multiplication

128 × 128 · 128 × 128 · f64 · 1 worker · allocated output · self product



How to read this report

Allocation-inclusive API latency unless marked reused output. Peer medians are observations.

Charts round to 6 significant digits (≈). The exact-value appendix preserves all recorded digits.

Kernix 0.12.0 / Sparse matrices & memory

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Sparse matrix × vector

Per-call median

2048 × 2048 · five diagonals · f64 · 1 worker · five diagonals · prepared operands · owned output · finite contracts differ



Sparse × dense matrix

Per-call median

512 × 512 · 32 RHS · five diagonals · f64 · 1 worker · five diagonals · prepared operands · owned output · finite contracts differ



Kernix 0.12.0 / Sparse matrices & memory

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Sparse × dense matrix

Per-call median

512 × 512 · 32 RHS · five diagonals · f32 · 1 worker · five diagonals · prepared operands · owned output · finite contracts differ



Sparse × sparse matrix

Per-call median

256 × 256 · five diagonals each · f64 · 1 worker · five diagonals · prepared operands · owned output · finite contracts differ



Kernix 0.12.0 / Sparse matrices & memory

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Automatic CSR construction

512 × 512 · clone + scan + compress · f64 · Construction · dense copy or compression · preparation costs included



Kernix 0.12.0 / Matrix multiplication

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication · transposed view

Per-call median

128 × 128 · 128 × 128 · f64 · 1 worker · borrowed transpose · allocated output · independent operands



Matrix multiplication · transposed view

Per-call median

128 × 128 · 128 × 128 · f32 · 1 worker · borrowed transpose · allocated output · independent operands



Kernix 0.12.0 / Matrix multiplication

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Independent matrix operands

Per-call median

128 × 128 · 128 × 128 · f64 · 1 worker · allocated output · independent operands



Independent matrix operands

Per-call median

128 × 128 · 128 × 128 · f32 · 1 worker · allocated output · independent operands



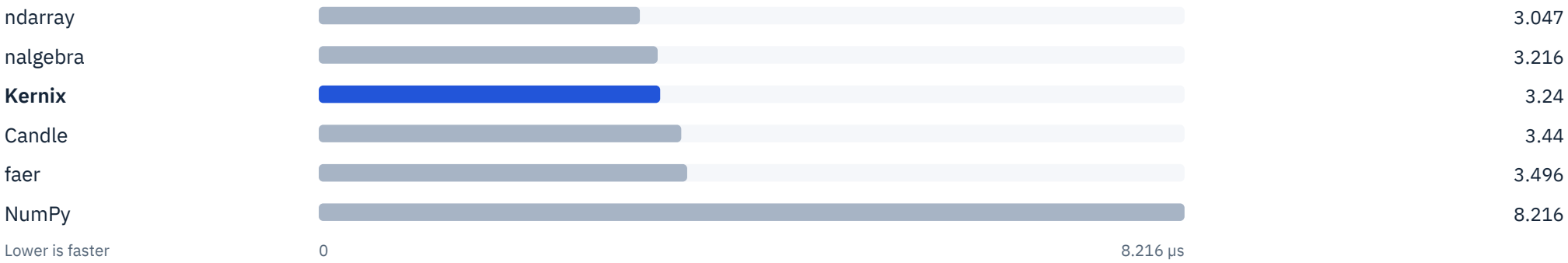
Kernix 0.12.0 / Matrix multiplication

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

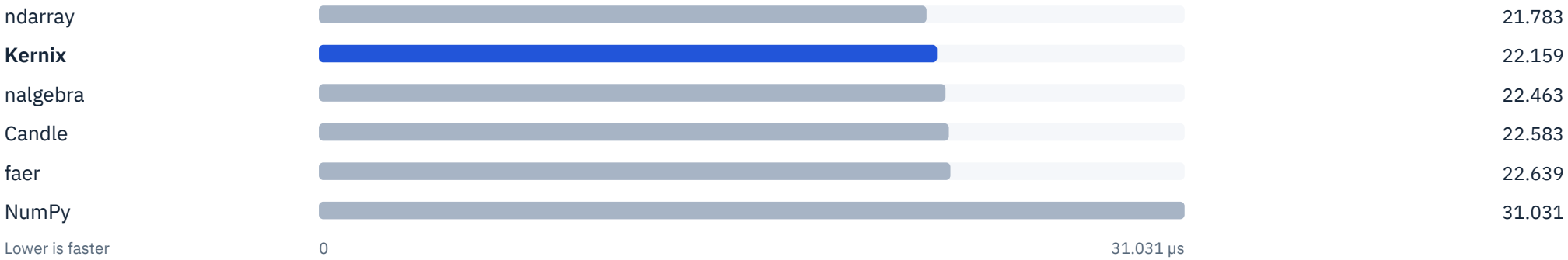
32 × 32 · 32 × 32 · f64 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f64 · 1 worker · allocated output · self product



Kernix 0.12.0 / Matrix multiplication

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

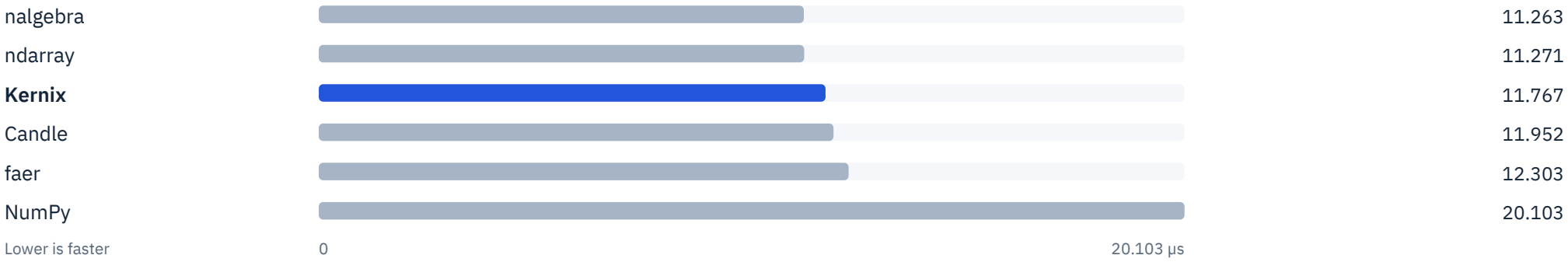
256 × 256 · 256 × 256 · f64 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f32 · 1 worker · allocated output · self product



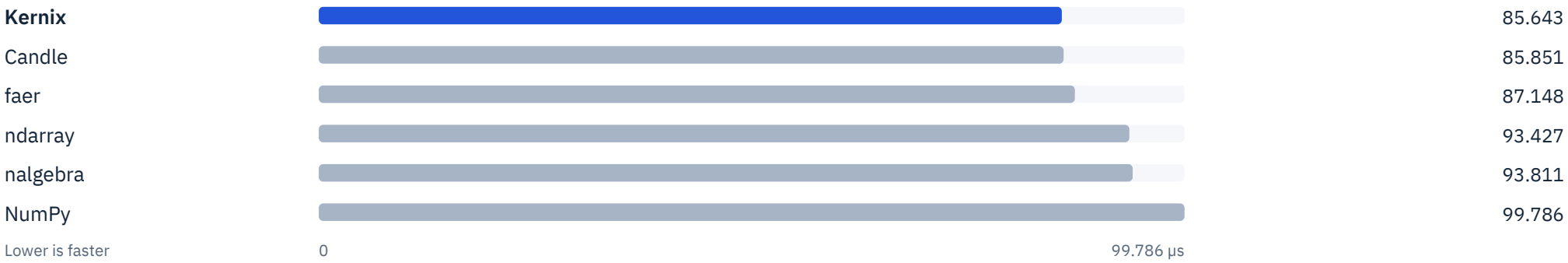
Kernix 0.12.0 / Matrix multiplication

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

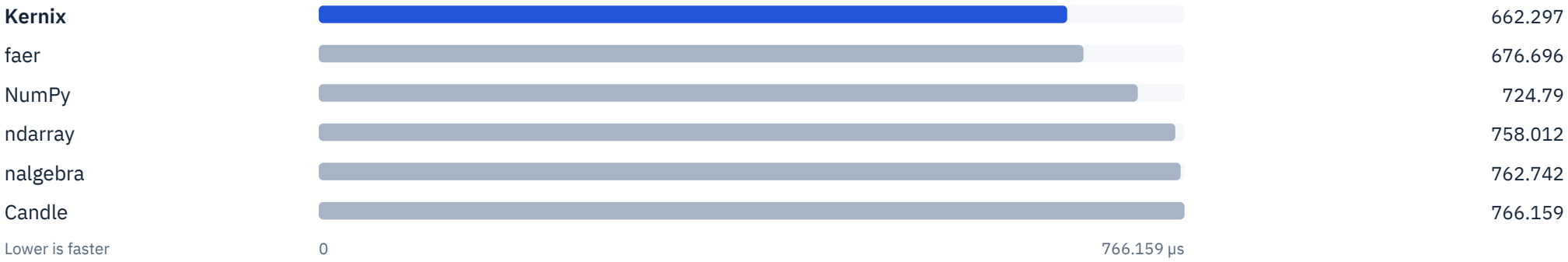
128 × 128 · 128 × 128 · f32 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

256 × 256 · 256 × 256 · f32 · 1 worker · allocated output · self product



Kernix 0.12.0 / Matrix multiplication

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Batched multiplication

Per-call median

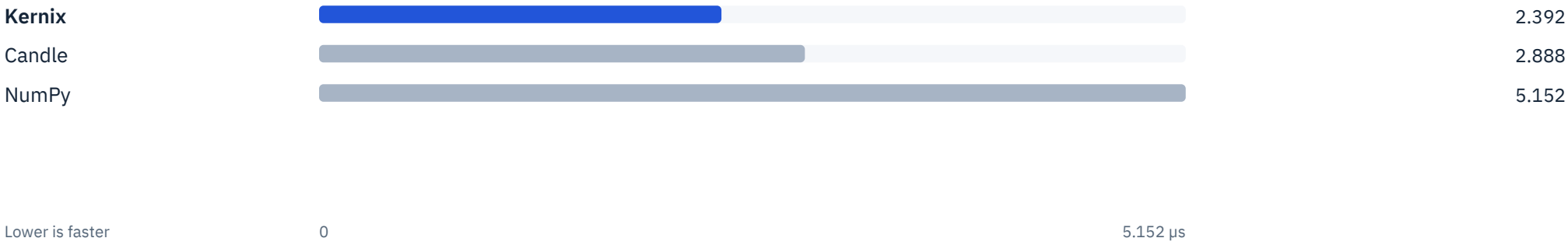
4 × 16 × 32 · 1 × 32 × 16 · f64 · 1 worker · allocated output · broadcast batches



Batched multiplication

Per-call median

4 × 16 × 32 · 1 × 32 × 16 · f32 · 1 worker · allocated output · broadcast batches



Kernix 0.12.0 / Explicit parallel execution

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Matmul · 2 workers · reused output

Per-call median

64 × 64 · 64 × 64 · f64 · 2 workers · reused output · pool preparation excluded



Matmul · 2 workers · reused output

Per-call median

128 × 128 · 128 × 128 · f64 · 2 workers · reused output · pool preparation excluded



Kernix 0.12.0 / Explicit parallel execution

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Matmul · 2 workers · reused output

Per-call median

256 × 256 · 256 × 256 · f64 · 2 workers · reused output · pool preparation excluded



Matmul · 2 workers · reused output

Per-call median

128 × 128 · 128 × 128 · f32 · 2 workers · reused output · pool preparation excluded



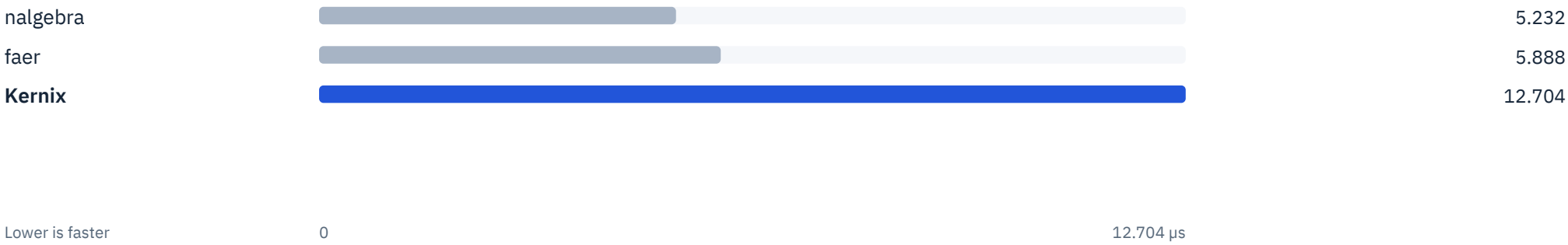
Kernix 0.12.0 / Decompositions & solvers

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky solve · reused factor

Per-call median

128 × 128, 1 RHS · f64 · 1 worker · prepared reusable factor · owned output · factorization excluded



Cholesky solve · reused factor

Per-call median

256 × 256, 64 RHS · f64 · 1 worker · prepared reusable factor · owned output · factorization excluded



Kernix 0.12.0 / Decompositions & solvers

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky solve · reused factor

Per-call median

512 × 512, 128 RHS · f64 · 1 worker · prepared reusable factor · owned output · factorization excluded



LU solve · reused factor

Per-call median

256 × 256, 64 RHS · f64 · 1 worker · prepared reusable factor · owned output · factorization excluded



Kernix 0.12.0 / Decompositions & solvers

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky rank-one update

256 × 256 · f64 · Prepared factors · owned output · refactor uses preformed modified matrix



Cholesky rank-one downdate

256 × 256 · f64 · Prepared factors · owned output · refactor uses preformed modified matrix



Kernix 0.12.0 / Decompositions & solvers

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

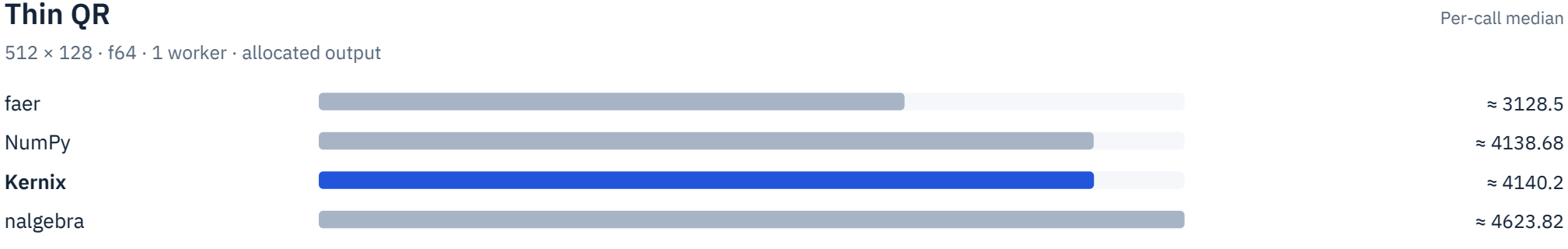
Cholesky rank-one update

512 × 512 · f64 · Prepared factors · owned output · refactor uses preformed modified matrix



Thin QR

512 × 128 · f64 · 1 worker · allocated output



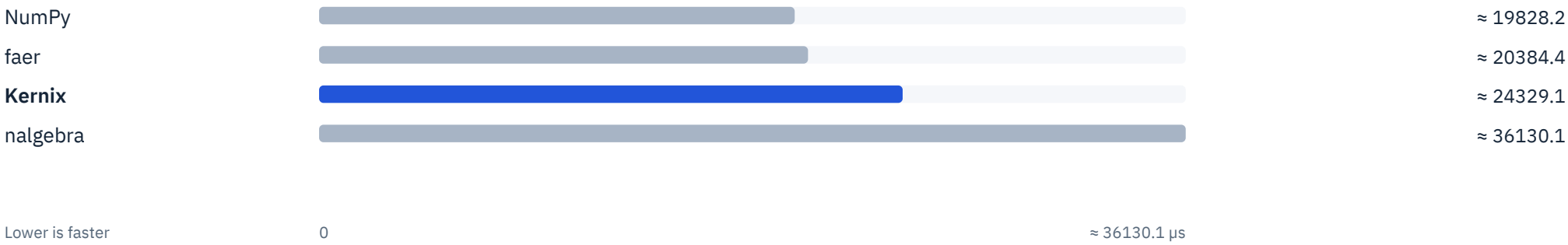
Kernix 0.12.0 / Decompositions & solvers

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Thin QR

1024 × 256 · f64 · 1 worker · allocated output

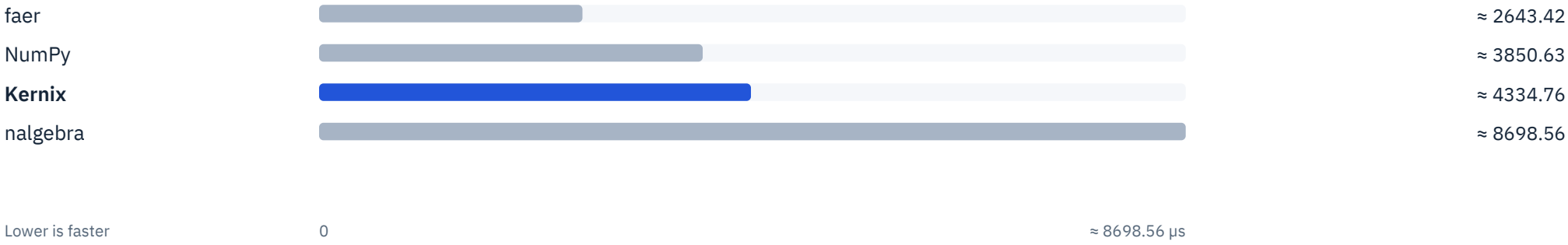
Per-call median



Cholesky factorization

512 × 512 · f64 · 1 worker · allocated output

Per-call median



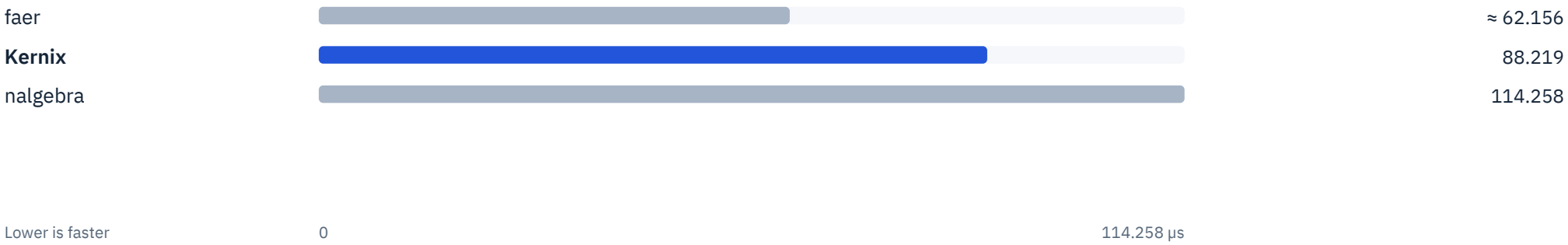
Kernix 0.12.0 / Decompositions & solvers

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Column-pivoted QR

96 × 32 · f64 · 1 worker · allocated output

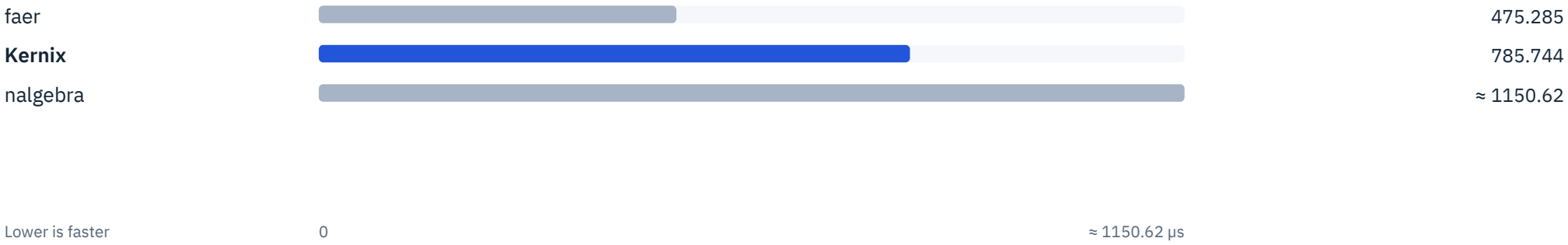
Per-call median



Column-pivoted QR

256 × 64 · f64 · 1 worker · allocated output

Per-call median



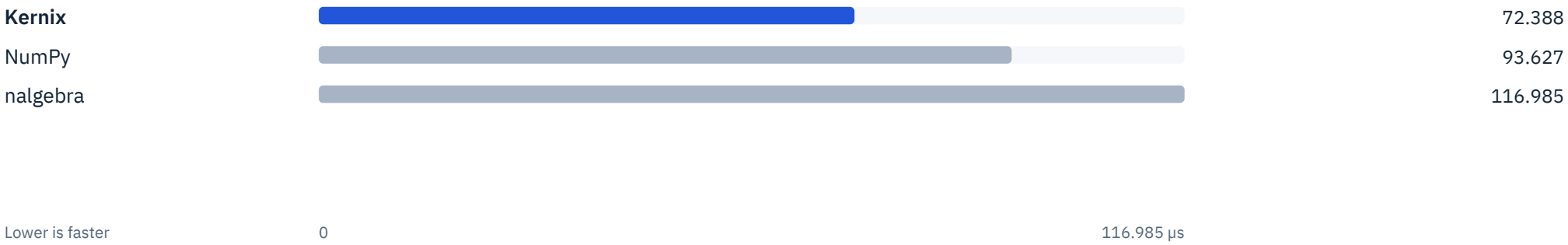
Kernix 0.12.0 / Decompositions & solvers

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Minimum-norm least squares

Per-call median

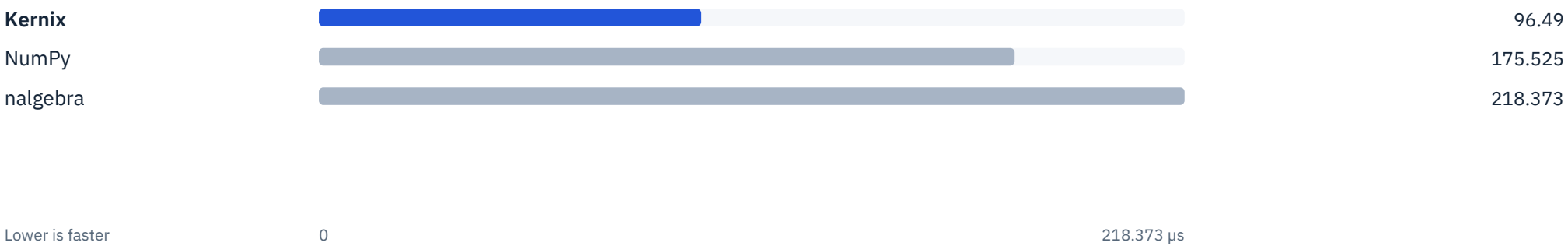
64 × 32, rank 16, 4 RHS · f64 · 1 worker · 4 RHS · factorization included · algorithms differ



Minimum-norm least squares

Per-call median

32 × 64, rank 32, 4 RHS · f64 · 1 worker · 4 RHS · factorization included · algorithms differ



Kernix 0.12.0 / Decompositions & solvers

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

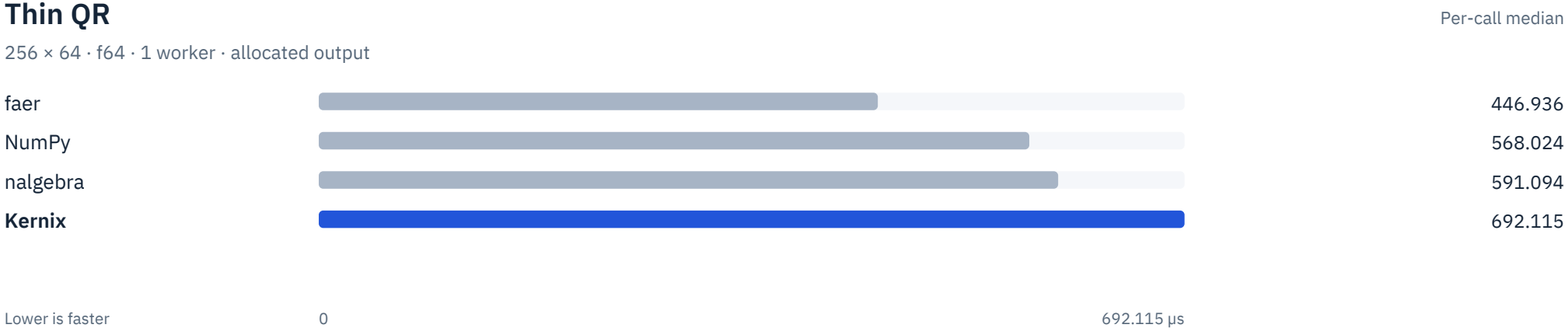
Thin QR

96 × 32 · f64 · 1 worker · allocated output



Thin QR

256 × 64 · f64 · 1 worker · allocated output



Kernix 0.12.0 / Decompositions & solvers

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky factorization

64 × 64 · f64 · 1 worker · allocated output



Cholesky factorization

128 × 128 · f64 · 1 worker · allocated output



Kernix 0.12.0 / Decompositions & solvers

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky factorization

256 × 256 · f64 · 1 worker · allocated output



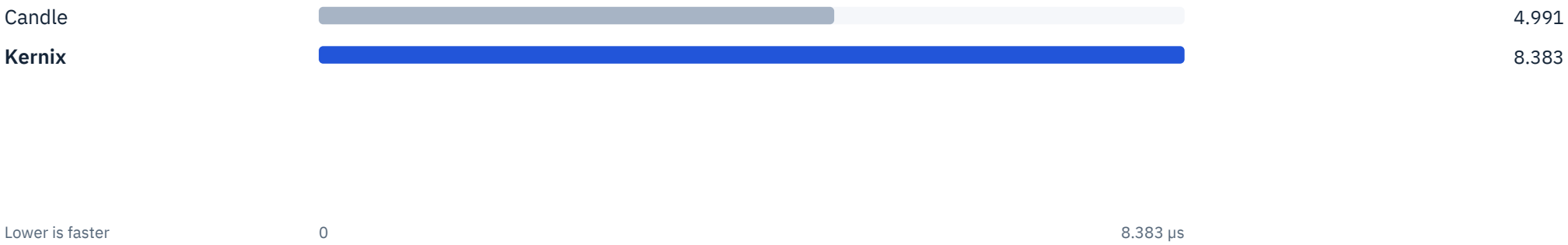
Kernix 0.12.0 / Training & autodiff

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Shared-variable backward

Per-call median

sum(X * X), 64 × 32; prepared graph · f64 · 1 worker · prepared graph · fresh gradients



Record + backward

Per-call median

sum(A · W), 32 × 16 · 16 × 8 · f64 · 1 worker · allocated output · self product



Kernix 0.12.0 / Training & autodiff

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Model inference

Per-call median

batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



Model recording + forward

Per-call median

batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



Kernix 0.12.0 / Training & autodiff

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Model backward

Per-call median

prepared graph → fresh gradients · f64 · 1 worker · prepared graph · fresh gradients



Parameter update

Per-call median

transactional SGD, fixed gradients · f64 · 1 worker · preparation as documented



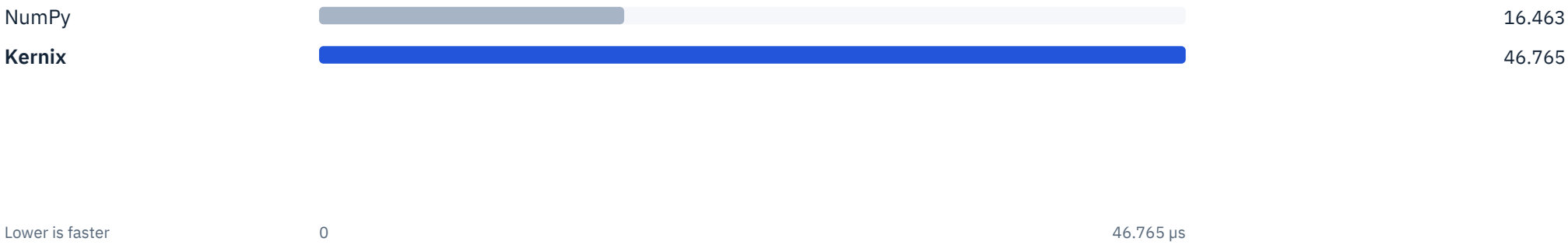
Kernix 0.12.0 / Tensor contractions

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Coalesced tensor contraction

16 × 32 × 8 · 32 × 8 × 16 · f64 · 1 worker · allocated output

Per-call median



Kernix 0.12.0 / Statistics & activations

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Row softmax

256 × 64 · f64 · 1 worker · allocated output

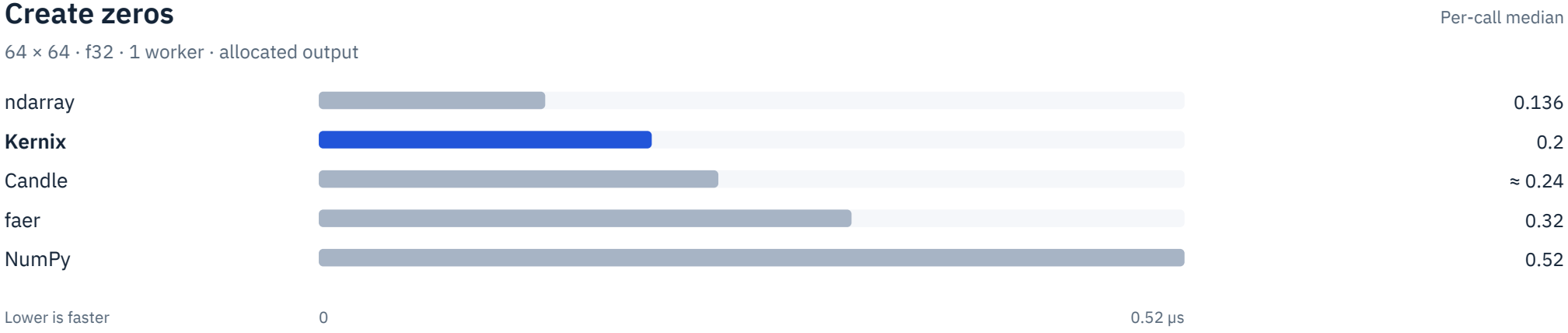


Kernix 0.12.0 / Tensor construction

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Create zeros

64 × 64 · f32 · 1 worker · allocated output



Create ones

64 × 64 · f32 · 1 worker · allocated output



Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Sparse matrix × vector / SciPy · CSR 2048 × 2048 · five diagonals · f64	9791	9.791
Sparse matrix × vector / NumPy 2048 × 2048 · five diagonals · f64	810426	810.426
Sparse matrix × vector / faer 2048 × 2048 · five diagonals · f64	8008	8.008
Sparse matrix × vector / Kernix · explicit dense 2048 × 2048 · five diagonals · f64	2947906	2947.906
Sparse × dense matrix / SciPy · CSR 512 × 512 · 32 RHS · five diagonals · f64	23031	23.031
Sparse × dense matrix / Kernix · explicit dense 512 × 512 · 32 RHS · five diagonals · f64	783359	783.359
Sparse × dense matrix / NumPy 512 × 512 · 32 RHS · five diagonals · f64	1070132	1070.132
Sparse × dense matrix / faer 512 × 512 · 32 RHS · five diagonals · f64	67549	67.549
Sparse × dense matrix / SciPy · CSR 512 × 512 · 32 RHS · five diagonals · f32	15407	15.407
Sparse × dense matrix / Kernix · explicit dense 512 × 512 · 32 RHS · five diagonals · f32	381241	381.241

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Sparse × dense matrix / faer 512 × 512 · 32 RHS · five diagonals · f32	65740	65.74
Sparse × dense matrix / NumPy 512 × 512 · 32 RHS · five diagonals · f32	510943	510.943
Sparse × sparse matrix / SciPy · CSR 256 × 256 · five diagonals each · f64	47413	47.413
Sparse × sparse matrix / NumPy 256 × 256 · five diagonals each · f64	1476689	1476.689
Sparse × sparse matrix / Kernix · explicit dense 256 × 256 · five diagonals each · f64	1324937	1324.937
Automatic CSR construction / NumPy 512 × 512 · clone + scan + compress · f64	55437	55.437
Automatic CSR construction / SciPy · CSR 512 × 512 · clone + scan + compress · f64	755079	755.079
Automatic CSR construction / Kernix · explicit dense 512 × 512 · clone + scan + compress · f64	54277	54.277
Matrix multiplication · transposed view / NumPy 128 × 128 · 128 × 128 · f64	196260	196.26
Matrix multiplication · transposed view / Kernix 128 × 128 · 128 × 128 · f64	210676	210.676

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication · transposed view / Kernix 128 × 128 · 128 × 128 · f32	92707	92.707
Matrix multiplication · transposed view / NumPy 128 × 128 · 128 × 128 · f32	100266	100.266
Independent matrix operands / NumPy 128 × 128 · 128 × 128 · f64	195437	195.437
Independent matrix operands / Kernix 128 × 128 · 128 × 128 · f64	168343	168.343
Independent matrix operands / Kernix 128 × 128 · 128 × 128 · f32	85651	85.651
Independent matrix operands / NumPy 128 × 128 · 128 × 128 · f32	100115	100.115
Matrix multiplication / faer 32 × 32 · 32 × 32 · f64	3496	3.496
Matrix multiplication / nalgebra 32 × 32 · 32 × 32 · f64	3216	3.216
Matrix multiplication / NumPy 32 × 32 · 32 × 32 · f64	8216	8.216
Matrix multiplication / ndarray 32 × 32 · 32 × 32 · f64	3047	3.047

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / Candle 32 × 32 · 32 × 32 · f64	3440	3.44
Matrix multiplication / Kernix 32 × 32 · 32 × 32 · f64	3240	3.24
Matrix multiplication / Candle 64 × 64 · 64 × 64 · f64	22583	22.583
Matrix multiplication / nalgebra 64 × 64 · 64 × 64 · f64	22463	22.463
Matrix multiplication / NumPy 64 × 64 · 64 × 64 · f64	31031	31.031
Matrix multiplication / Kernix 64 × 64 · 64 × 64 · f64	22159	22.159
Matrix multiplication / faer 64 × 64 · 64 × 64 · f64	22639	22.639
Matrix multiplication / ndarray 64 × 64 · 64 × 64 · f64	21783	21.783
Matrix multiplication / faer 128 × 128 · 128 × 128 · f64	168974	168.974
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f64	209459	209.459

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f64	194213	194.213
Matrix multiplication / nalgebra 128 × 128 · 128 × 128 · f64	209517	209.517
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f64	168207	168.207
Matrix multiplication / Candle 128 × 128 · 128 × 128 · f64	169255	169.255
Matrix multiplication / NumPy 256 × 256 · 256 × 256 · f64	1470099	1470.099
Matrix multiplication / Candle 256 × 256 · 256 × 256 · f64	1322686	1322.686
Matrix multiplication / nalgebra 256 × 256 · 256 × 256 · f64	1331869	1331.869
Matrix multiplication / Kernix 256 × 256 · 256 × 256 · f64	1323125	1323.125
Matrix multiplication / ndarray 256 × 256 · 256 × 256 · f64	1323957	1323.957
Matrix multiplication / faer 256 × 256 · 256 × 256 · f64	1540097	1540.097

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / Kernix 64 × 64 · 64 × 64 · f32	11767	11.767
Matrix multiplication / Candle 64 × 64 · 64 × 64 · f32	11952	11.952
Matrix multiplication / NumPy 64 × 64 · 64 × 64 · f32	20103	20.103
Matrix multiplication / nalgebra 64 × 64 · 64 × 64 · f32	11263	11.263
Matrix multiplication / faer 64 × 64 · 64 × 64 · f32	12303	12.303
Matrix multiplication / ndarray 64 × 64 · 64 × 64 · f32	11271	11.271
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f32	85643	85.643
Matrix multiplication / Candle 128 × 128 · 128 × 128 · f32	85851	85.851
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f32	99786	99.786
Matrix multiplication / faer 128 × 128 · 128 × 128 · f32	87148	87.148

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / nalgebra 128 × 128 · 128 × 128 · f32	93811	93.811
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f32	93427	93.427
Matrix multiplication / faer 256 × 256 · 256 × 256 · f32	676696	676.696
Matrix multiplication / Kernix 256 × 256 · 256 × 256 · f32	662297	662.297
Matrix multiplication / ndarray 256 × 256 · 256 × 256 · f32	758012	758.012
Matrix multiplication / nalgebra 256 × 256 · 256 × 256 · f32	762742	762.742
Matrix multiplication / NumPy 256 × 256 · 256 × 256 · f32	724790	724.79
Matrix multiplication / Candle 256 × 256 · 256 × 256 · f32	766159	766.159
Batched multiplication / NumPy 4 × 16 × 32 · 1 × 32 × 16 · f64	12991	12.991
Batched multiplication / Candle 4 × 16 × 32 · 1 × 32 × 16 · f64	4423	4.423

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Batched multiplication / Kernix 4 × 16 × 32 · 1 × 32 × 16 · f64	3824	3.824
Batched multiplication / Candle 4 × 16 × 32 · 1 × 32 × 16 · f32	2888	2.888
Batched multiplication / Kernix 4 × 16 × 32 · 1 × 32 × 16 · f32	2392	2.392
Batched multiplication / NumPy 4 × 16 × 32 · 1 × 32 × 16 · f32	5152	5.152
Matmul · 2 workers · reused output / Kernix 64 × 64 · 64 × 64 · f64	16439	16.439
Matmul · 2 workers · reused output / Kernix 128 × 128 · 128 × 128 · f64	93139	93.139
Matmul · 2 workers · reused output / Kernix 256 × 256 · 256 × 256 · f64	695435	695.435
Matmul · 2 workers · reused output / Kernix 128 × 128 · 128 × 128 · f32	48558	48.558
Cholesky solve · reused factor / nalgebra 128 × 128, 1 RHS · f64	5232	5.232
Cholesky solve · reused factor / Kernix 128 × 128, 1 RHS · f64	12704	12.704

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Cholesky solve · reused factor / faer 128 × 128, 1 RHS · f64	5888	5.888
Cholesky solve · reused factor / faer 256 × 256, 64 RHS · f64	449476	449.476
Cholesky solve · reused factor / nalgebra 256 × 256, 64 RHS · f64	1293099	1293.099
Cholesky solve · reused factor / Kernix 256 × 256, 64 RHS · f64	1718431	1718.431
Cholesky solve · reused factor / nalgebra 512 × 512, 128 RHS · f64	12891520	12891.52
Cholesky solve · reused factor / faer 512 × 512, 128 RHS · f64	3194277	3194.277
Cholesky solve · reused factor / Kernix 512 × 512, 128 RHS · f64	13545170	13545.17
LU solve · reused factor / nalgebra 256 × 256, 64 RHS · f64	1231346	1231.346
LU solve · reused factor / faer 256 × 256, 64 RHS · f64	455592	455.592
LU solve · reused factor / Kernix 256 × 256, 64 RHS · f64	1654184	1654.184

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Cholesky rank-one update / Kernix refactor 256 × 256 · f64	771917	771.917
Cholesky rank-one update / nalgebra 256 × 256 · f64	34007	34.007
Cholesky rank-one update / Kernix 256 × 256 · f64	151832	151.832
Cholesky rank-one downdate / Kernix refactor 256 × 256 · f64	767346	767.346
Cholesky rank-one downdate / nalgebra 256 × 256 · f64	33982	33.982
Cholesky rank-one downdate / Kernix 256 × 256 · f64	150079	150.079
Cholesky rank-one update / Kernix refactor 512 × 512 · f64	4349557	4349.557
Cholesky rank-one update / Kernix 512 × 512 · f64	651662	651.662
Cholesky rank-one update / nalgebra 512 × 512 · f64	147704	147.704
Thin QR / Kernix 512 × 128 · f64	4140200.9999999995	4140.2009999999995

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Thin QR / NumPy 512 × 128 · f64	4138682	4138.682
Thin QR / faer 512 × 128 · f64	3128496	3128.496
Thin QR / nalgebra 512 × 128 · f64	4623818	4623.818
Thin QR / nalgebra 1024 × 256 · f64	36130138	36130.138
Thin QR / Kernix 1024 × 256 · f64	24329143	24329.143
Thin QR / faer 1024 × 256 · f64	20384433	20384.433
Thin QR / NumPy 1024 × 256 · f64	19828189	19828.189
Cholesky factorization / faer 512 × 512 · f64	2643417	2643.417
Cholesky factorization / Kernix 512 × 512 · f64	4334758	4334.758
Cholesky factorization / nalgebra 512 × 512 · f64	8698559	8698.559

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Cholesky factorization / NumPy 512 × 512 · f64	3850626	3850.626
Column-pivoted QR / faer 96 × 32 · f64	62156.00000000001	62.15600000000001
Column-pivoted QR / Kernix 96 × 32 · f64	88219	88.219
Column-pivoted QR / nalgebra 96 × 32 · f64	114258	114.258
Column-pivoted QR / Kernix 256 × 64 · f64	785744	785.744
Column-pivoted QR / faer 256 × 64 · f64	475285	475.285
Column-pivoted QR / nalgebra 256 × 64 · f64	1150615	1150.615
Minimum-norm least squares / Kernix 64 × 32, rank 16, 4 RHS · f64	72388	72.388
Minimum-norm least squares / NumPy 64 × 32, rank 16, 4 RHS · f64	93627	93.627
Minimum-norm least squares / nalgebra 64 × 32, rank 16, 4 RHS · f64	116985	116.985

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Minimum-norm least squares / NumPy 32 × 64, rank 32, 4 RHS · f64	175525	175.525
Minimum-norm least squares / Kernix 32 × 64, rank 32, 4 RHS · f64	96490	96.49
Minimum-norm least squares / nalgebra 32 × 64, rank 32, 4 RHS · f64	218373	218.373
Thin QR / Kernix 96 × 32 · f64	73299	73.299
Thin QR / faer 96 × 32 · f64	65140	65.14
Thin QR / NumPy 96 × 32 · f64	66372	66.372
Thin QR / nalgebra 96 × 32 · f64	61309.00000000001	61.30900000000001
Thin QR / nalgebra 256 × 64 · f64	591094	591.094
Thin QR / NumPy 256 × 64 · f64	568024	568.024
Thin QR / faer 256 × 64 · f64	446936	446.936

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Thin QR / Kernix 256 × 64 · f64	692115	692.115
Cholesky factorization / NumPy 64 × 64 · f64	17695	17.695
Cholesky factorization / Kernix 64 × 64 · f64	27311	27.311
Cholesky factorization / nalgebra 64 × 64 · f64	20830	20.83
Cholesky factorization / faer 64 × 64 · f64	13223	13.223
Cholesky factorization / nalgebra 128 × 128 · f64	130271.99999999999	130.27199999999999
Cholesky factorization / NumPy 128 × 128 · f64	85684	85.684
Cholesky factorization / faer 128 × 128 · f64	65092.99999999999	65.09299999999999
Cholesky factorization / Kernix 128 × 128 · f64	276018	276.018
Cholesky factorization / faer 256 × 256 · f64	407338	407.338

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Cholesky factorization / nalgebra 256 × 256 · f64	1033901	1033.901
Cholesky factorization / NumPy 256 × 256 · f64	748579	748.579
Cholesky factorization / Kernix 256 × 256 · f64	766542	766.542
Shared-variable backward / Kernix sum(X * X), 64 × 32; prepared graph · f64	8383	8.383
Shared-variable backward / Candle sum(X * X), 64 × 32; prepared graph · f64	4991	4.991
Record + backward / Kernix sum(A · W), 32 × 16 · 16 × 8 · f64	3503	3.503
Record + backward / Candle sum(A · W), 32 × 16 · 16 × 8 · f64	7216	7.216
Model inference / Candle batched normalized/sliced MLP · f64	5352	5.352
Model inference / Kernix batched normalized/sliced MLP · f64	3039	3.039
Model recording + forward / Candle batched normalized/sliced MLP · f64	7615	7.615

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Model recording + forward / Kernix batched normalized/sliced MLP · f64	3808	3.808
Model backward / Kernix prepared graph → fresh gradients · f64	4168	4.168
Model backward / Candle prepared graph → fresh gradients · f64	21230	21.23
Parameter update / Kernix transactional SGD, fixed gradients · f64	728	0.728
Coalesced tensor contraction / Kernix 16 × 32 × 8 · 32 × 8 × 16 · f64	46765	46.765
Coalesced tensor contraction / NumPy 16 × 32 × 8 · 32 × 8 × 16 · f64	16463	16.463
Row softmax / Candle 256 × 64 · f64	110298	110.298
Row softmax / Kernix 256 × 64 · f64	113186	113.186
Row softmax / NumPy 256 × 64 · f64	134585	134.585
Row softmax / ndarray 256 × 64 · f64	97291	97.291

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Create zeros / NumPy 64 × 64 · f32	520	0.52
Create zeros / ndarray 64 × 64 · f32	136	0.136
Create zeros / faer 64 × 64 · f32	320	0.32
Create zeros / Candle 64 × 64 · f32	239.99999999999997	0.23999999999999997
Create zeros / Kernix 64 × 64 · f32	200	0.2
Create ones / Candle 64 × 64 · f32	344	0.344
Create ones / faer 64 × 64 · f32	239.99999999999997	0.23999999999999997
Create ones / ndarray 64 × 64 · f32	208	0.208
Create ones / Kernix 64 × 64 · f32	256	0.256
Create ones / NumPy 64 × 64 · f32	1544	1.544